

Dietary polyamine intake and risk of colorectal adenomatous polyps

Background: Putrescine, spermidine, and spermine are the polyamines required for human cell growth. The inhibition of ornithine decarboxylase (ODC), which is the rate-limiting enzyme of polyamine biosynthesis, decreases tumor growth and the development of colorectal adenomas. A database was developed to estimate dietary polyamine exposure and relate exposure to health outcomes. Objective: We hypothesized that high polyamine intake would increase risk of colorectal adenoma and that the allelic variation at ODC G>A +316 would modify the association. Design: Polyamine exposure was estimated in subjects pooled (n = 1164) from the control arms of 2 randomized trials for colorectal adenoma prevention [Wheat Bran Fiber low-fiber diet arm (n = 585) and Ursodeoxycholic Acid placebo arm (n = 579)] by using baseline food-frequency questionnaire data. All subjects had to have a diagnosis of colorectal adenoma to be eligible for the trial. Results: A dietary intake of polyamines above the median amount in the study population was associated with 39% increased risk of colorectal adenoma at follow-up (adjusted OR: 1.39; 95% CI: 1.06, 1.83) in the pooled sample. In addition, younger participants (OR: 1.94; 95% CI: 1.23, 3.08), women (OR: 2.43; 95% CI: 1.48, 4.00), and ODC GG genotype carriers (OR: 1.59; 95% CI: 1.00, 2.53) had significantly increased odds of colorectal adenoma if they consumed above-median polyamine amounts. Conclusions: This study showed a role for dietary polyamines in colorectal adenoma risk. Corroboration of these findings would confirm a previously unrecognized, modifiable dietary risk factor for colorectal adenoma.